Arista Access Node BAS-ACC-78 High-availability Failure Resolution SOP for Hardware/Environmental Faults (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8935

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for resolving hardware/environmental faults leading to ASIC memory parity errors, line card crashes, and power supply redundancy loss on the Arista Access Node BAS-ACC-78 at the Basavanagudi POP. The Arista Access Node is a critical component of the network infrastructure, providing high-speed, secure access to network resources. The hardware and environmental failures described in this SOP can have significant impacts on network availability and performance. Understanding the underlying hardware mechanics and protocols is essential to resolving these issues quickly and effectively. The Arista Access Node's architecture is based on a modular design, with multiple line cards and power supplies working in tandem to provide high availability. However, hardware and environmental failures can still occur, causing ASIC memory parity errors, line card crashes, or power supply redundancy loss. These failures can be detected by the system's monitoring and alarm systems, triggering the ALARM_8935 alarm code and severity warning. To resolve these issues, it is essential to follow a systematic approach, starting with diagnostic commands to identify the root cause, followed by resolution steps to fix the issue, and finally verification commands to ensure the fix is successful. This SOP provides a detailed guide for resolving hardware/environmental faults on the Arista Access Node BAS-ACC-78, ensuring high availability and minimizing downtime.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Basavanagudi POP

- Verify the redundant link status between the Arista Access Node BAS-ACC-78 and the adjacent network equipment

- Check the environmental conditions at the Basavanagudi POP, including temperature, humidity, and power supply redundancy

Potential Root Causes

Cause	Probability
ASIC memory parity error due to defective memory or faulty memory controller	High
Line card crash due to software or hardware failure	Medium
Power supply redundancy loss due to failure of primary or secondary power supply	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal memory, Expected Output: Memory usage and parity error information, Description: This command displays memory usage and parity error information, helping to identify if the ASIC memory parity error is due to defective memory or a faulty memory controller.

Command: show system internal linecard, Expected Output: Line card status and error information, Description: This command displays line card status and error information, helping to identify if the line card crash is due to software or hardware failure.

Command: show power supply, Expected Output: Power supply status and redundancy information, Description: This command displays power supply status and redundancy information, helping to identify if the power supply redundancy loss is due to failure of primary or secondary power supply.
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Resolution Steps

Step 1: Step: Replace defective memory or faulty memory controller, Command: clear system internal memory, Description: This command clears the memory parity error and resets the memory controller, resolving the ASIC memory parity error.

Step 2: Step: Replace faulty line card, Command: no system internal linecard, Description: This command removes the faulty line card and replaces it with a new one, resolving the line card crash.

Step 3: Step: Replace failed power supply, Command: set power supply redundancy, Description: This command configures the power supply redundancy, ensuring that the system has a backup power supply in case of failure.

Post-Check Verification

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Command: show system internal memory, Expected Output: Memory usage and parity error information (resolved), Description: This command verifies that the memory parity error has been resolved.

Command: show system internal linecard, Expected Output: Line card status and error information (resolved), Description: This command verifies that the line card crash has been resolved.

Command: show power supply, Expected Output: Power supply status and redundancy information (resolved), Description: This command verifies that the power supply redundancy loss has been resolved.
```

Roll-back Procedures

- Step: Restore original configuration, Command: show running-config, Description: This command restores the original configuration in case the fix fails.

- Step: Reset system, Command: reset, Description: This command resets the system to its original state in case the fix fails.